



Agent goes on a cooling spree and hunts down elusive aftermarket CPU and GPU cooling solutions

small fan to cool these chips. Nearly all high-performance, low latency RAM also comes with aluminium heatspreaders to dissipate heat produced in the memory chips. However the heat produced in CPUs and graphics cards is much more than the heat produced in memory and motherboards. Especially if you opt for a high-end configuration, stock cooling may not cut it.

Since the processor and graphics solutions are by far the biggest culprits when it comes to producing heat, I'm going to cover solutions available for cooling them down. You might ask yourself – why should I spend 2000 bucks on an aftermarket cooler when I already have a cooler with my CPU / attached to my graphics card? Quite simply, these coolers are designed to be just about adequate. Moreover, the geographical regions where companies like Intel, AMD and NVIDIA test the efficiency of such coolers are generally much cooler

The real chill factor

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If you're one of those people who've been following technology for the last six years or so the term room heater would mean more than just the electrical device we use to lessen the severity of winter indoors. Critics used to refer to Intel's Pentium 4 processors by this nickname owing to the extraordinary temperatures they would reach. Another infamous component was NVIDIA's GeForce 5800 Ultra, fondly called the dust buster because of the whirring sound its fan produced when spinning at full tilt.

While stock cooling does a

fair job of keeping entry-level and mid-range components cool, even these components benefit from after-market cooling solutions with lower temperatures, longer working life spans and less noise. Remember, heat is the universal enemy of all integrated circuitry and an increase in temperature of 10 degrees Celsius can bring about a 25 per cent decrease in the total life of the product, and this applies to CPUs, graphics cards, RAM, HDDs and just about any IC.

In India, an additional enemy is dust. Dust reduces working efficiency of heatsinks, fans and just about any cooling solution. Even a liquid cooling solution experiences a decrease in working efficiency due to dust build up on

the radiator block and fan over time. Fan bearings gradually wear down because of dust and their rpm slowly decreases – and this happens irrespective of how good the fan. I've actually experimented with measuring the rpm of a new fan and again measuring it after six months sans maintenance and was surprised to see its speed reduced by 15 per cent, and I'm talking about a good fan I bought for 500 bucks – the average 100 rupee fan will likely conk off in under a year. Obviously, the more powerful the system you build the more cooling and ventilation it requires. Cooling is also dependent on cabinet ventilation; after all fans need to suck air from the environment and blow it on to hot components.

than India and such solutions are rarely adjusted for the tropics. In cases where your CPU or GPU is producing more heat than the thermal solution can dissipate, an aftermarket solution will control temperatures. Previously, aftermarket coolers were a rarity in India but this has changed for CPU coolers at least.

The concept of using heatpipes to whisk away heat is also common, and many aftermarket solutions utilise heatpipes embedded amid the fins. Remember from your physics that the larger the surface area of metal in contact with air the better the cooling; therefore the larger the heatsink and the more fins the better cooling it will provide, although a lot also